System Capacity & Care Load Analytics

HHS Unaccompanied Alien Children Program

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Executive Summary

Overall System Status: **HIGH**

Significant concern. Activate contingency plans and increase discharge processing capacity.

Dataset Overview

Dataset Overview

Metric	Value
Total Records	1075
Date Range Start	2023-01-12 00:00:00
Date Range End	2025-12-21 00:00:00
Total Columns	53

Data Columns

Column	Data Type
Date	datetime64[ns]
CBP_Apprehended	float64
CBP_Custody	float64
Transfers	float64
HHS_Care	float64
Discharges	float64
Total_System_Load	float64
Net_Daily_Intake	float64
Daily_Growth_Rate	float64
Backlog_Indicator	float64
Discharge_Offset_Ratio	float64
CBP_Custody_Rolling_Mean_7d	float64
CBP_Custody_Rolling_Std_7d	float64
CBP_Custody_Rolling_Var_7d	float64
Transfers_Rolling_Mean_7d	float64
Transfers_Rolling_Std_7d	float64
Transfers_Rolling_Var_7d	float64
HHS_Care_Rolling_Mean_7d	float64
HHS_Care_Rolling_Std_7d	float64
HHS_Care_Rolling_Var_7d	float64
Discharges_Rolling_Mean_7d	float64
Discharges_Rolling_Std_7d	float64
Discharges_Rolling_Var_7d	float64
Total_System_Load_Rolling_Mean_7d	float64
Total_System_Load_Rolling_Std_7d	float64

Column	Data Type
Total_System_Load_Rolling_Var_7d	float64
Net_Daily_Intake_Rolling_Mean_7d	float64
Net_Daily_Intake_Rolling_Std_7d	float64
Net_Daily_Intake_Rolling_Var_7d	float64
CBP_Custody_Rolling_Mean_14d	float64
CBP_Custody_Rolling_Std_14d	float64
CBP_Custody_Rolling_Var_14d	float64
Transfers_Rolling_Mean_14d	float64
Transfers_Rolling_Std_14d	float64
Transfers_Rolling_Var_14d	float64
HHS_Care_Rolling_Mean_14d	float64
HHS_Care_Rolling_Std_14d	float64
HHS_Care_Rolling_Var_14d	float64
Discharges_Rolling_Mean_14d	float64
Discharges_Rolling_Std_14d	float64
Discharges_Rolling_Var_14d	float64
Total_System_Load_Rolling_Mean_14d	float64
Total_System_Load_Rolling_Std_14d	float64
Total_System_Load_Rolling_Var_14d	float64
Net_Daily_Intake_Rolling_Mean_14d	float64
Net_Daily_Intake_Rolling_Std_14d	float64
Net_Daily_Intake_Rolling_Var_14d	float64
Week_Number	UInt32
Month	int32
Quarter	int32
Year	int32
Day_of_Week	int32
Weekend_Flag	bool

Data Validation Report

Data Validation Summary

Overall Validation Status: **FAIL**

Failed Checks:

- transfers_le_cbp

Key Performance Indicators

KPI	Value
Total Children Under Care	2484.0
Average Daily Intake	12.260465116279069
Average Daily Discharges	11.873488372093023
Average Length of Stay	167.48186046511628
Average Bed Occupancy	608.1283720930232
Net Intake Rate	0.3825581395348837
Care Load Volatility	0.4663152841971363
Backlog Accumulation	1.3891892551021042
Discharge Efficiency	0.9638118957714767
Average System Load	0.2450765581395349

Healthcare Analytics Summary

Overall System Status: **HIGH**

Recommendation: Significant concern. Activate contingency plans and increase discharge processing capacity.

Warnings:

- Capacity stress detected

- Backlog increasing

- High net intake pressure

Key Analytics Results

Metric	Value
Capacity Stress Status	MODERATE_STRESS
Stress Days	140
Backlog Trend	INCREASING
Current Backlog	12232
Net Intake Trend	INCREASING
Growth Status	DECLINE
Total Growth %	-62.2%

Recommendations

Significant concern. Activate contingency plans and increase discharge processing capacity.

Specific Actions:

1. Increase discharge processing capacity and review placement procedures to reduce backlog accumulation.

Conclusion

This report provides a comprehensive analysis of the HHS Unaccompanied Alien Children Program capacity and care load metrics. The current system status is **HIGH**. The analysis includes data validation, feature engineering, KPI calculations, and specialized healthcare analytics covering capacity stress, backlog trends, pressure windows, and growth patterns. Continuous monitoring and regular analysis of these metrics will help ensure optimal system performance and enable proactive response to capacity challenges.